

Paper 30.25 - Grand Ribbon Toolkit Supplement

CQE-paper-30.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-30.25.md

Paper 30.25 - Grand Ribbon Toolkit Supplement

Purpose

This supplement shows how to reproduce Paper 30's corpus ribbon. The proof is in Paper 30 and its formal verifier. The toolkit exposes the repeated eight-slot form so a reader can inspect the corpus without mistaking the framing for a new theorem.

Digital Route

Run:

```
`python production/formal-papers/CQE-paper-30/verify_grand_ribbon_meta_framer.py`
```

The expected status is ``pass_with_open_packaging_obligations``. The verifier checks the slot schema, fills papers 00-29 as a 30-position sweep, builds a terminal composition route, checks the transport-obligation ledger, and confirms that Paper 31 is prepared only as readout.

Analog Route

Draw one long ribbon with eight bins per position:

```
`C L R B T O W A`
```

Create positions for papers 00 through 29. At each position, place a center card, a left-boundary card, a right-boundary card, a boundary-rule card, a tool/receipt card, an obligation card, a workbook card, and an anchor card.

A bin is closed only when it has both a value and a provenance tag. If the value exists without provenance, mark it as open. If an obligation exists, place it in the ``O`` bin instead of removing it.

Hidden-Guess Diagnostic

Training mode should hide the label until after the reviewer chooses one:

- filled ribbon slot,
- unfilled ribbon slot,
- valid provenance row,
- open transport lift,
- invalid hidden lift,
- valid Paper 31 readout,
- invalid Paper 31 dependency.

The answer key distinguishes a proof-stack position from a retrospective readout.